Game Theory: A Very Short Introduction

ECON 1071: Incentives

September 11, 2023

Plan For Today

- Short motivation
- Normal-form games: definitions
- NE in pure strategies + examples
- Mixed strategies, NE in mixed strategies + examples

Motivation 1

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- Gaming industry
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Terrorist

Security		Attack Terminal A	Attack Terminal B
	Guard Terminal A	1, -1	-1, 1
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Figure 2. Attack Terminal A or Terminal B using Values

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Many many non-trivial insights about might happen in all of the above, using the *same* set of basic tools.

Motivation 2

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> *"I do not think that the bargaining models that I myself have studied have significant predictive value... I am captivated by the charm of formal models: tales emerge from the formal symbols, and these tales have almost miraculous powers over me. On the other hand, I am obsessively occupied with* **denying any interpretation contending that economic models produce conclusions of real value***" (Rubinstein, Economic Fables)*

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- We take the complete opposite approach in this class.
- But generally always good to keep this in mind, and take everything with a grain of salt.

Example

https://www.youtube.com/watch?v=yM38mRHY150

Takeways

- Ingredients: players, payoffs, actions, "rules."
- Given the above, we try to predict a reasonable outcome.
- But: Set of reasonable outcomes the model predicts can be large.
- GT can't always help us rule out possibilities.

Formal Definitions

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1. A finite set of **players**: $N = \{1, 2, \ldots, n\}$
2. A set of (pure) **strategies** S_i for each player $i \in N$.
   - Induces a set of **strategy profiles**: $S = S_1 \times \cdots \times S_n$.
3. A set of **utility functions** $u_i : S \to \mathbb{R}$ for each player $i \in N$.

Interpretation

1. A finite set of players N (usually $N = \{1, 2, \ldots, n\}$)
   - This just defines who participates in the game.
2. A set of (pure) *strategies* S_i for each $i \in N$.
   - S_i is the set of actions that player i can take in the game.
3. A set of *utility functions* $u_i : S \to \mathbb{R}$ for each $i \in N$.
   - When each player i chooses an action $s_i \in S_i$ simultaneously, we get a strategy profile $s = (s_1, s_2, ..., s_n)$.
   - Each such profile induces a payoff for each player i, denoted $u_i(S)$
     $\hookrightarrow$ *What reward player i gets depends on what the other players do*
     - Sometimes convenient to write as $u_i(s_i, s_{-i})$
     - s_{-i} is conventional notation for "the profile of strategies of everyone but i"

Split-or-Steal Example

- $N = \{\textbf{L}\text{ucy},\ \textbf{T}\text{ony}\}$
- $S_{\text{L}} = S_{\text{T}} = \{\textbf{SP}\text{lit},\textbf{ST}\text{eal}\}$

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- $S_{\text{L}} = S_{\text{T}} = \{\textbf{SP}\text{lit},\textbf{ST}\text{eal}\}$
- $$u_L(s_L, s_T) = \begin{cases} 66,885 & s_L = \text{ST}, s_T = \text{SP} \\ 33,442.5 & s_L = \text{SP}, s_T = \text{SP} \\ 0 & s_L \in \{ST, SP\}, s_T = \text{ST} \end{cases}$$
- u_T is symmetric.

Matrix Representation

Two-player games with finite strategies are usually represented as a **payoff matrix**:

		Tony	
		SP	**ST**
Lucy	**SP**	33, 33	0, 66
	ST	66, 0	0, 0

Nash Equilibrium: Motivation

We want to be able to make a reasonable prediction

- We think of outcomes as possible strategy profiles (not payoffs!)
- One possible definition of "reasonable" is a notion of rationality or stability
- "Everyone optimizes w.r.t. to others' behavior"
- "No one can gain from deviating"

Pure-strategy Nash Equilibrium

Nash equilibrium = "*No player can profitably deviate, fixing everybody else's strategy*"

Definition (Pure strategy Nash equilibrium)

In a normal-form game $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$, a **pure-strategy Nash equilibrium** is a pure-strategy profile s^* such that for every player i,

$$\underbrace{u_i\left(s_i^* s_{-i}^*\right)}_{\text{payoff } i \text{ gets for sticking to } s^*} \geq \underbrace{u_i\left(s_i', s_{-i}^*\right)}_{\text{payoff } i \text{ gets for switching to } s_i'} \quad \text{for all } s_i' \in S_i.$$

Back to Lucy and Tony

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- (**ST**, **ST**) NE: No player can profitably deviate

Lucy and Tony Revisited

- So NE is "anything but an equal split."
- Not ruling out too much!
- We assumed Lucy and Tony only care about payoffs.
- But people have feelings!

Fairness/Regret

Let's refine our model to incorporate **regret**.

- Assume both feel slightly worse when the opponent steals their money (relative to mutual stealing).
- Could capture inequity aversion ("why do they get the money and I don't?"), regret ("why did I trust them?") etc.
- Parameterize regret by ε:

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- Q: Does the set of NE change?
- A: for any $\varepsilon > 0$, the unique NE is $(\mathbf{ST}, \mathbf{ST})$

Guilt

Now let's instead refine our model to incorporate **guilt**.

- Guilt = "I feel bad for stealing, since the other player put split..."
- Parametrize guilt by γ:

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- Q: Does the set of NE change?
- A:
  - if $\gamma > 33$, the set of equilibria is $\{(\mathbf{ST}, \mathbf{ST}), (\mathbf{SP}, \mathbf{SP})\}$.
  - Otherwise, the set of equilibria is as in the original game $\{(\mathbf{ST}, \mathbf{SP}), (\mathbf{SP}, \mathbf{ST}), (\mathbf{ST}, \mathbf{ST})\}$.

Finding pure-strategy NE

- Equivalent to finding a fixed point (difficult problem!)
- With a finite number of strategies, usually brute force works (esp. with two players).

Ice cream on a linear beach

What if the strategy space is not finite?

Exercise: write the following as a normal-form game.

- Two ice cream vendors are on a beach that stretches the interval $[0, 1]$.
- Customers are uniformly distributed along that interval.
- Customers go to the closest vendor.
- If the vendors choose an identical position, customers choose at random.

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- $S_1 = S_2 = [0, 1]$ (each picks a spot in $[0, 1]$ to locate shop)
- $u_i(s_i, s_j) = \begin{cases} \frac{s_i + s_j}{2} & s_i < s_j \\ 1 - \frac{s_i + s_j}{2} & s_i > s_j \\ \frac{1}{2} & s_i = s_j \end{cases}$

Ice cream on a linear beach: Equilibrium

Claim: Unique NE is $(s_i^*, s_j^*) = (\frac{1}{2}, \frac{1}{2})$. Steps for proving this:

1. If vendor i chooses $s_i = \frac{1}{2}$, $u_i(\frac{1}{2}, s_j) \geq \frac{1}{2}$ (easy to verify)
2. In equilibrium we must have $u_1, u_2 \geq \frac{1}{2}$
   - Assume not, then by #1 one player has a profitable deviation, a contradiction.
3. $u_1 = u_2 = \frac{1}{2}$
   - By #2 + the fact that $u_1 + u_2 = 1$ by the definition of the game.
4. $s_i^* = s_j^* = \frac{1}{2}$
   - Assume $s_j^* < \frac{1}{2}$, then for $s_i' = s_j + \varepsilon$ $(\varepsilon \to 0)$ $u_i(s_i', s_j^*) \approx 1 - s_j^* > \frac{1}{2}$, a contradiction to #3.
   - Symmetric argument for $s_j^* > \frac{1}{2}$.

$\hookrightarrow$ *Has many important applications (e.g. Downsian convergence in elections).*

Rock Paper Scissors

p2

		R	**P**	**S**
	R	0, 0	-1, 1	1, -1
p1	**P**	1, -1	0, 0	-1, 1
	S	-1, 1	1, -1	0, 0

- Q: Nash equilibrium in pure strategies?

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- Q: Nash equilibrium in pure strategies?
- A: No pure-strategy NE exists!

Mixed-strategy NE

- As example above shows, NE in pure strategies need not exist.
- Does this mean we can't make predictions?
- How do *you* play RPS?
- If I draw 200 players from the population, what distribution of play should I expect to find?

Mixed strategies

Definition (Probability distribution)

A **probability distribution** over a set X is a function $f : X \to [0, 1]$ such that $\sum_{x \in X} f(x) = 1$.

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Definition (Mixed strategy)

Let $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ be a normal-form game. A **mixed strategy** for player i, denoted σ_i, is a probability distribution over S_i.

$\hookrightarrow$ *Interpretation: Player i randomly picks a strategy from S_i, according to some probability distribution σ_i. They pick s_i with probability $\sigma_i(s_i)$.*

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Notation: mixed strategy that puts probability p_k on strategy $s_i^{(k)} \in S_i$ is written as $p_1 s_i^{(1)} \oplus p_2 s_i^{(2)} \oplus \cdots \oplus p_K s_i^{(K)}$.

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Pure strategies are also mixed strategies (w/ degenerate distributions).

Computing utilities of mixed strategies

- Recall: with pure strategies, player i's utility $u_i(s_i, s_{-i})$ depends on the entire strategy profile $s = (s_i, s_{-i})$.
- Let $\sigma = (\sigma_1, \ldots, \sigma_n)$ be a **profile of mixed strategies**.
  $\hookrightarrow$ induces a distribution over S.
- Player i's utility from the mixed strategy profile σ is given by

$$u_i(\sigma) = \sum_{s \in S} \underbrace{\sigma_1(s_1) \ldots \sigma_n(s_n)}_{\text{probability } s \text{ is played}} u_i(s) = \mathbb{E}_{s \in S}[u_i(S)].$$

- Also convenient to write as

$$u_i(\sigma_i, \sigma_{-i}) = \sum_{s_i \in S_i} \sigma_i(s_i)\, u_i(s_i, \sigma_{-i})$$

$\hookrightarrow$ $u_i(\sigma_i, \sigma_{-i})$ is a weighted average of $u_i(s_i, \sigma_{-i})$'s

Nash Equilibrium in Mixed Strategies

Nash equilibrium = *"No player can profitably deviate, fixing everybody else's strategy"*

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In a normal-form game $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ a mixed-strategy Nash equilibrium is a mixed strategy profile σ^* such that for every player i,

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- Q: Why not require $u_i\left(\sigma_i^*, \sigma_{-i}^*\right) \geq u_i\left(\sigma_i', \sigma_{-i}^*\right)$ for all $\sigma_i' \in \Delta S_i$
  - ($\Delta(X)$ is set of distributions over X)

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  - ($\Delta(X)$ is set of distributions over X)
- A: b/c $u_i(\sigma_i', \sigma_{-i})$ is simply a weighted average of $u_i(s, \sigma_{-i})$'s, so $u_i\left(\sigma_i^*, \sigma_{-i}^*\right)$ is also larger than all convex combos of $u_i(s, \sigma_{-i})$.

Useful Facts about NE in mixed strategies

1. With finite # of players, finite # of strategies per player, always exists.
   - Existence proof follows more or less directly from Kakutani's fixed-point theorem (sorry)
   - May not exist if set of strategies is non-compact, e.g. "the player that chooses the higher real number wins"
2. If σ^* is a Nash equilibrium, then for s_i and s_i' such that $\sigma_i^*(s_i) > 0$ and $\sigma_i^*(s_i') > 0$, we have $u_i\left(s_i, \sigma_{-i}^*\right) = u_i\left(s_i', \sigma_{-i}^*\right)$
   - All strategies that a player plays with strictly positive probabilities give the same utility, given others' play σ_{-i}.

Proof of Useful Fact #2

- To arrive at a contradiction assume that $u_i\left(s_i, \sigma^*_{-i}\right) \neq u_i\left(s'_i, \sigma^*_{-i}\right)$ for some s_i and s'_i such that $\sigma^*_i\left(s_i\right) > 0$ and $\sigma^*_i\left(s'_i\right) > 0$.
- Take $\hat{s} \equiv \underset{s \in S_i, \sigma_i(s) > 0}{\arg\max} \left\{u_i\left(s, \sigma^*_{-i}\right)\right\}$.
  - i.e., player i's strategy that produces highest utility (conditional on others playing σ^*_{-i}) out of all those that are played by player i with strictly positive probability.
- Since $u_i\left(\sigma^*_i, \sigma^*_{-i}\right)$ is a weighted average, we must have $u_i\left(\hat{s}, \sigma^*_{-i}\right) > u_i\left(\sigma^*_i, \sigma^*_{-i}\right)$
  - Because by the contradiction assumption $u_i\left(\sigma^*_i, \sigma^*_{-i}\right)$ averages over elements that are strictly smaller than $u_i\left(\hat{s}, \sigma^*_{-i}\right)$.
- But this contradicts the definition of σ^*_i being a part of a NE. QED

How to find NE in mixed strategies?

- Generally, very hard problem to solve.
- Simple brute force will generally never work—we can't just go over all possible distributions.
- Below we show two specific examples that make use of Useful Fact #2
- This and similar methods can be used when the number of strategies is relatively small.
- And in general Useful Fact #2 is indeed extremely useful in those proofs.
- But no general recipe: involves educated guess work + intuition, then verifying our hopefully correct guesses.

Back to RPS

Claim: The unique NE is $\frac{1}{3}R \oplus \frac{1}{3}P \oplus \frac{1}{3}S, \frac{1}{3}R \oplus \frac{1}{3}P \oplus \frac{1}{3}S$.

- **Both players mix:** Assume only one, then mixing is not a best response for the other. A contradiction.
- **Strictly positive probabilities on R, P, S:** Assume e.g. $\sigma_i(S) = 0$, then j's best response is P. A contradiction since can't be degenerate.
- By useful fact #2, $u_i(R, \sigma_j) = u_i(P, \sigma_j) = u_i(S, \sigma_j)$.
- Solve 3 equations w/ 3 unknowns, get $\sigma_j(R) = \sigma_j(P) = \sigma_j(S) = \frac{1}{3}$. Use symmetry to show

Game of assurance

	L	R
T	1,1	0,0
B	0,0	2,2

- Pure-strategy NE are easy to find: $(T, L), (B, R)$.

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- Simple arguments show that if one player mixes, both do.
- Assume player 1 plays T w/ Pr p. By Useful Fact #2:

$$u(L, \sigma_1^*) = u(R, \sigma_1^*)$$

$$pu_2(L, T) + (1-p)u_2(L, B) = pu_2(R, T) + (1-p)u_2(R, B)$$

$$p \cdot 1 + (1-p) \cdot 0 = p \cdot 0 + (1-p) \cdot 2$$

$$3p = 2$$

$$p = \frac{2}{3}$$

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$$p \cdot 1 + (1-p) \cdot 0 = p \cdot 0 + (1-p) \cdot 2$$
$$3p = 2$$
$$p = \frac{2}{3}$$

- Use symmetry to show unique strictly mixed NE is $\frac{2}{3}T \oplus \frac{1}{3}B, \frac{2}{3}L \oplus \frac{1}{3}R$

Game of assurance

So the 3 equilibria are $(B, R), (T, L), (\frac{2}{3}T \oplus \frac{1}{3}B, \frac{2}{3}L \oplus \frac{1}{3}R)$.
What about payoffs?

- Payoffs under each of the 3 equilibria are $(2, 2), (1, 1), (\frac{2}{3}, \frac{2}{3})$.
- Why mixed NE gives lower payoff?
- Because strategies are independent of each other, we often hit outcomes that yield 0 for both players.